

AI INTERVIEW SYSTEM

MAJOR PROJECT SYNOPSIS

Submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology

in

Computer Science & Engineering – AIML

Submitted to

Department of Computer Science & Engineering

Lloyd Institute of Engineering & Technology

Plot No. 3, Knowledge Park II, Greater Noida, Uttar Pradesh 201306

Session: 2026-27



Submitted by

Stud Name: Rohit Kumar
Roll No: 2401720109022
Section: C

Under the Supervision of

Guide Name
Assistant Professor

INDEX

Sr. No.	Topic	Page No.
01	Introduction	1 - 2 Pages
02	Statement of Problem with Objective	1 Page
03	Literature Survey	1 Page
04	Expected Outcome / Scope of the Project	1 Page
05	Tentative Work Plan (mapped to the 10-week timeline)	1 Page
06	Software/Hardware required for the development of the project	1 Page
07	References	1 Page

Student's Signature

Guide's Signature

1. INTRODUCTION

The recruitment process is one of the most critical activities in any organization. Traditional recruitment methods involve manual resume screening, candidate shortlisting, conducting interviews, evaluating responses, and maintaining candidate records. These activities are often time-consuming, repetitive, and prone to human bias. With the rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Web Technologies, recruitment processes can be automated to improve efficiency and accuracy. The **AI Interview System** is a web-based intelligent recruitment platform designed to automate resume analysis, candidate assessment, interview management, and performance evaluation. The system assists recruiters in identifying suitable candidates by automatically extracting skills from uploaded resumes, generating interview questions based on the extracted skills, recording candidate interviews, and evaluating candidate performance.

The proposed system uses Resume Parsing techniques to extract textual information from uploaded resumes in PDF format. After extracting the resume content, the system identifies technical skills possessed by the candidate and generates relevant interview questions. Candidates can then participate in a mock interview where their responses are recorded and stored for evaluation.

The system also provides a dashboard for recruiters to monitor candidate performance, interview status, scores, and feedback. By integrating automation into the recruitment workflow, organizations can reduce hiring time, improve candidate screening quality, and maintain structured recruitment records. **Project Domain**

- Artificial Intelligence
- Human Resource Management
- Recruitment Automation
- Web Application Development

Technologies Used

- React.js
- Node.js
- Express.js
- PostgreSQL
- JavaScript
- Axios
- Multer
- PDF-Parser
- HTML5 & CSS3

Key Features

- Resume Upload and Parsing
- Skill Extraction
- Automated Question Generation
- Mock Interview System
- Video Recording

- Candidate Evaluation
- Dashboard and Analytics
- Interview Scheduling

The AI Interview System aims to provide a scalable and efficient recruitment platform that enhances the hiring process through automation and intelligent decision-making.

2. STATEMENT OF PROBLEM AND OBJECTIVES

Statement of Problem

In many organizations, recruitment activities are performed manually. HR personnel spend significant time reviewing resumes, identifying candidate skills, preparing interview questions, conducting interviews, and maintaining records. As the number of applicants increases, managing recruitment becomes increasingly difficult.

Some major challenges include:

- Large volume of resumes to review manually.
- Inconsistent evaluation of candidates.
- Human bias during screening.
- Time-consuming interview processes.
- Difficulty in tracking candidate performance.
- Lack of centralized recruitment data.

To overcome these limitations, an automated recruitment system is required that can intelligently analyze resumes, generate interview questions, evaluate candidates, and maintain recruitment records.

Objectives

The primary objectives of the AI Interview System are:

1. To automate resume screening and skill extraction.
2. To generate interview questions based on candidate skills.
3. To conduct mock interviews through a web platform.
4. To record interview sessions for future evaluation.
5. To evaluate candidate performance automatically.
6. To provide recruiters with candidate scores and feedback.
7. To maintain centralized recruitment records.
8. To reduce recruitment time and effort.
9. To improve hiring accuracy.
10. To provide a scalable recruitment solution for organizations.

3. LITERATURE SURVEY

A review of existing research and recruitment systems was conducted to understand current methodologies and identify gaps.

S.No	Paper Title	Authors	Year	Key Findings
1	AI-Based Recruitment Systems	Sharma et al.	2021	Demonstrated how AI improves candidate screening efficiency.
2	Automated Resume Parsing Techniques	Gupta et al.	2020	Discussed extraction of candidate information from resumes.
3	Interview Analytics Using Machine Learning	Kumar et al.	2022	Explained candidate evaluation using automated analysis.
4	Intelligent Human Resource Management Systems	Verma et al.	2023	Proposed automation in recruitment workflows.

Research Gap

Existing systems generally focus on only one aspect of recruitment, such as resume screening or interview management. Few systems provide a complete solution integrating:

- Resume Analysis
- Skill Extraction
- Question Generation
- Interview Recording
- Candidate Evaluation
- Performance Dashboard

The proposed AI Interview System combines all these functionalities into a single integrated platform.

4. EXPECTED OUTCOME / SCOPE OF THE PROJECT

Expected Outcome

The proposed system is expected to:

- Automate recruitment processes.
- Reduce manual effort in candidate screening.
- Improve recruitment efficiency.
- Generate relevant interview questions.
- Record interview sessions.
- Evaluate candidate performance objectively.
- Maintain centralized recruitment records.
- Assist recruiters in making better hiring decisions.

Scope of the Project

The scope of the AI Interview System includes:

Candidate Module

- Resume Upload
- Profile Management
- Interview Participation

Recruitment Module

- Candidate Management
- Question Generation
- Interview Scheduling

Evaluation Module

- Candidate Scoring
- Feedback Generation
- Performance Tracking

Administrative Module

- Dashboard
- Analytics
- Report Generation

Future Scope

Future enhancements may include:

- Machine Learning-Based Evaluation
- Emotion Detection
- Facial Expression Analysis
- Voice Sentiment Analysis
- AI Chatbot Interviewer
- Cloud Deployment
- Mobile Application Support

5. TENTATIVE WORK PLAN (10-WEEK TIMELINE)

Week Activity

Week 1 Requirement Analysis and Project Planning

Week 2 Literature Survey and Feasibility Study

Week 3 Database Design and System Architecture

Week 4 Resume Upload and Parsing Module

Week 5 Skill Extraction Module Development

Week 6 Interview Question Generation Module

Week 7 Mock Interview and Recording Module

Week 8 Candidate Evaluation and Scoring Module

Week 9 Dashboard Development and Testing

Week 10 Documentation, Report Writing and Final Presentation

Milestones

- Completion of Resume Parser
- Completion of Question Generator
- Completion of Interview Module
- Completion of Evaluation Module
- System Testing
- Final Deployment

6. SOFTWARE / HARDWARE REQUIRED

Software Requirements Operating System

- Windows 10 / Windows 11
- Linux (Optional)

Programming Languages

- JavaScript
- SQL

Frontend Technologies

- React.js
- HTML5
- CSS3

Backend Technologies

- Node.js
- Express.js

Database Development Tools

- PostgreSQL
- Visual Studio Code
- Git & GitHub
- Postman

Libraries Used

- Axios
- Multer
- PDF-Parse
- React Router

Hardware Requirements Minimum Configuration

- Processor: Intel Core i3
- RAM: 4 GB
- Storage: 100 GB
- Webcam
- Microphone
- Internet Connection

Recommended Configuration

- Processor: Intel Core i5/i7
- RAM: 8 GB or higher
- Storage: 256 GB SSD
- HD Webcam
- Noise-Cancelling Microphone

7. REFERENCES

1. React Documentation, Meta Platforms Inc., Available: <https://react.dev>
2. Node.js Documentation, OpenJS Foundation, Available: <https://nodejs.org>
3. PostgreSQL Documentation, PostgreSQL Global Development Group, Available: <https://www.postgresql.org>
4. Express.js Documentation, Available: <https://expressjs.com>
5. Mozilla Developer Network (MDN), JavaScript Documentation, Available: <https://developer.mozilla.org>
6. Sharma, R., Gupta, A., "AI-Based Recruitment Systems," International Journal of Computer Applications, 2021.
7. Kumar, P., Verma, S., "Interview Analytics Using Machine Learning," IEEE Conference Proceedings, 2022.
8. Gupta, N., "Automated Resume Parsing Techniques," Springer Journal of Information Systems, 2020.
9. Human Resource Management Review, Elsevier Publications.
10. Research articles related to Artificial Intelligence, Recruitment Automation, and Candidate Evaluation Systems.